SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 1 – SHORT ANSWER TEST (5 QUESTIONS)

Course Code:	1TA0509	**Course Title:**	**Computer Vision**
CO Assessed:	CO1	**Assessment:**	Assessment 1: Short Answer Test
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.***		
Student Name:	M. Neela Meghana	**Reg. No.:**	192311002
Section / Batch:	BE-CSE [Final year]	**Date:**	09/10/26

Code of Conduct

I M. Neela Meghana 192311002 *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student:

QUESTIONS: 5 **Total Marks:50**

Annexure A

Short Answer Test

Q1. Smart Traffic Monitoring System

A smart city is planning to install cameras at major traffic junctions to monitor vehicles, pedestrians, and traffic conditions. The captured images may vary in resolution, lighting, and camera position. **Apply the fundamental concepts of Computer Vision to design a suitable processing pipeline for this traffic monitoring system. Explain the major stages from image acquisition to object detection and classification.**

Q2. Automated Attendance System

A college wants to implement an automated attendance system using cameras that capture students sitting inside classrooms. The images may contain different lighting conditions, face orientations, shadows, and multiple students.

Analyze how image representation, preprocessing, feature extraction, and face recognition can be applied to develop the proposed attendance system. Discuss the major challenges that may affect its accuracy.

Q3. Medical Image Analysis System

A hospital plans to use a Computer Vision system to assist doctors in analyzing X-ray images for identifying abnormal regions. The images may contain noise, poor contrast, and different brightness levels.

Evaluate the importance of image resolution, intensity, preprocessing, segmentation, and feature extraction in this application. Explain how these factors can influence the accuracy and reliability of the system.

Q4. Security Surveillance System

A shopping mall wants to develop an intelligent surveillance system using cameras installed at entrances, corridors, and parking areas. The system must identify people and objects under different lighting and crowded conditions.

Analyze how the fundamental stages of a Computer Vision system can be used for this application.

Explain the effects of illumination, occlusion, camera position, and image quality on the final results.

Q5. Smart Agriculture Monitoring

A farming organization wants to use drone cameras to identify healthy plants, damaged plants, weeds, and disease-affected areas in large agricultural fields. The captured images may contain shadows, different lighting conditions, and variations in plant size.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, in-depth4, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature:	Signature: _____	Signature: _____
Date: 26/10/26.	Date: _____	Date: _____

1. Traffic Monitoring System Pipeline.

[Acquisition] → [Preprocessing] → [Segmentation] → [Detection]

[classification] ↵

1. Image Acquistion : Cameras capture live video streams. Images are stored as 3D matrices $I(x,y,c)$ with Red, Green, and Blue channels.

2. Preprocessing :

Resizing : Standardises dimensions (e.g., 640 x 640) using bilinear interpolation.

Enhancement : Uses CLAHE to fix headlight glare & night shadows.

Denoising : Applies a Gaussian filter to smooth out sensor noise.

3. Segmentation : Uses Mixture of Gaussians (MoG2) background subtraction to separate moving vehicles from the static road background.

4. Object Detection : Employs an anchor-box network to extract spatial edges and draw boundings boxes around objects.

5. Classification : Passes features into a softmax layer to assign final labels (Car, Bus, Pedestrians).

2. Automated Attendance System.

1. Core Technical Stages

- Representation : Captured color images are converted to 8-bit grayscale matrices ($Y = 0.299R + 0.587G + 0.114B$) to speed up Processing
- Preprocessing : Uses CLAHE to balance uneven window lighting and applies affine transformations to align tilted faces.

- Feature Extraction : Deep networks (like ArcFace) extract a 512-dimensional vector embedding of facial geometry (eye distance, nose bridge).
- Recognition : Matches live embeddings against a student database using Cosine Similarity.

2. Major challenges & Fixes

- Face Orientations : students looking away. Fix : Train with multi-angles.
- Bad Lightning : Dark shadows in back rows.
  Fix : Deploy Infrared (IR) sensors.
- Occlusion : Students blocking each other.
  Fix : Mount cameras at an elevated overhead angle.

3. Medical Image Analysis System (X-Ray)

[High Res / 16-bit Input] → [Median Filter + CLAHE]
→ [Otsu / U-Net Segment] → [GLCM Extraction]

- Image Resolution : High pixel density is mandatory to detect micro-fractures and tiny early-stage lesions.
- Intensity : Requires 12-bit (or) 16-bit depth to capture subtle density differences between bone and soft tissue.
- Preprocessing : Uses a Median Filter to remove sensor noise without blurring structural bone edges, followed by CLAHE to boost contrast.
- segmentation : Isolates specifies organs or bones using otsu Thresholding or U Net architectures to separate healthy tissue from abnormalities.
- Feature Extraction : Extracts texture metrics to define lesion boundaries, directly dictating clinical diagnostic reliability

ity Surveillance System :

1. Fundamental stages

- Acquisition : Continuous video capture across varied mall environments.
- Preprocessing : Dynamic contrast adjustment to handle intense sunlight at glass entrances.
- Segmentation : Using models like Mask R-CNN to separate human silhouettes from crowded backdrops.

2. Environmental Degradation Effects.

- Illumination : Moving between bright doors & dark corridors alters color profiles, causing tracking models to fail.
- Occlusion : Crowds temporarily block targets, breaking continuous tracking lines and swapping identities.

Smart Agriculture Monitoring

Multispectral Capture → Homomorphic Filter → Otsu Segment

SVM classify ← NDVI Extraction ←

1. Image Acquisition : Uses drone cameras to capture visible RGB and Near-Infrared light bands.

   Purpose : Collects raw health data invisible to human eyes.

2. Preprocessing : Applies Homomorphic Filtering to remove cloud shadows and orthorectifies images.

   Purpose : Standardises scale & lighting across fields.

3. Image Segmentation : Uses Excess Green and Otsu thresholding. Isolates green plant leaves by removing soil and rock background data.

4. Feature Extraction : Calculates NDVI pixels :

$$\frac{NIR - Red.}{NIR + Red.}$$

Purpose : Quantifies chlorophyll health levels and leaf textures.

5. Classification : Runs data through a Random Forest or SVM classifier. Purpose : Generates a map dividing fields into Healthy, Weeds or Diseased zones.